

सेंट्रल ट्रांसमिशन यूटिलिटी ऑफ इंडिया लिमिटेड

(पावर ग्रिड कॉर्पोरेशन ऑफ इंडिया लिमिटेड के स्वामित्व में)

(भारत सरकार का उदयम)

CENTRAL TRANSMISSION UTILITY OF INDIA LTD.

(A wholly owned subsidiary of Power Grid Corporation of India Limited)

(A Government of India Enterprise)

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Subject: 14th Consultation Meeting for Evolving Transmission Schemes in Northern Region-Minutes of Meeting

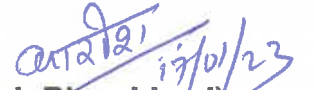
Dear Sir/Ma'am,

Please find enclosed the minutes of the 14th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 23rd December, 2022 through virtual mode.

The minutes are also available at CTU website (www.ctuil.in)

Thanking you,

Yours faithfully,



(Kashish Bhambhani)
General Manager (CTU)

Distribution List:

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Distribution List: Connectivity/Long-term Access Applicants

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Minutes of 14th Consultation Meeting for Evolving Transmission Schemes in Northern Region held on 23.12.2022

The 14th Consultation Meeting for Evolving Transmission Schemes in Northern Region (CMETS-NR) was held through VC on 23/12/2022. List of participants is enclosed at **Annexure-X**.

1. Confirmation on the minutes of the 13th CMETS-NR meeting

It was deliberated that the minutes of the 13th CMETS-NR meeting held on 30/11/2022 were issued vide letter dated 21/12/2022. However, it was informed that in the minutes circulated, for grant of St-II Connectivity to M/s TP Saurya Limited for 300 MW at Bikaner-III PS, the transmission system required for St-II Connectivity was inadvertently noted through 220 kV S/c (high capacity) line on D/c tower, however, “high capacity” line shall not be required in the instant case. Considering above, intimation for grant of St-II Connectivity was issued. Other than above, no comments were received, accordingly, minutes were confirmed as circulated.

Further, it was informed that Hon’ble CERC has notified the CERC (Connectivity and General Network Access (GNA) to the Inter-State Transmission System) Regulations, 2022. The GNA Regulations have come into operation with effect from 15-10-2022 except for provisions of Regulations 23 to 24, 26 to 36, 37.9, 38, 40 and 43, whose date of commencement shall be notified separately. However, notification for submission of fresh applications for Connectivity and GNA and their processing and grant of GNA Regulation had not been received by CTU.

Accordingly, it was informed that receipt of applications through NSWS portal under extant Connectivity Regulations has been resumed w.e.f. 00 hours of 01.12.2022. In view of this, St-I, St-II & LTA applications received in the month of Sep’22, Oct’22 & Nov’22 are being taken up for discussion in the present meeting. The detailed agenda was discussed as below:

A. Application related matters in Northern Region

2. Proposal for grant of Connectivity to the applications received from Renewable Energy Sources

It was informed that following Stage-I Connectivity applications were received in the month of Sep’22, Oct’22 & Nov’22 and were agreed to be granted as per the details below:

TABLE 1

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
St-I Applications received in the month of September'22								
1.	0112100011	Seven Renewable Power Private Limited	Bikaner distt., Rajasthan	02.09.2022	30.06.2024/ 900 Revised Start Date: 30/01/2025	Generator (Solar)	Bhadla-III PS	Seven Renewable Power Private Limited Solar Power Project – Bhadla-III PS 400 kV S/c line (Suitable to carry minimum 900 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
2.	0112100014	Tepsol Sun Sparkle Private Limited	Jaisalmer distt., Rajasthan	12.09.2022	31.03.2025/ 1000	Renewable Power Park Developer (Solar)	Bhadla-III PS	Tepsol Sun Sparkle Private Limited Renewable Power Park Project – Bhadla-III PS 400 kV S/c line (high capacity) (Suitable to carry minimum 1000 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
3.	0112100013	ReNew Dinkar Urja Private Limited	Bikaner distt., Rajasthan	13.09.2022	01.10.2023/ 200 Revised Start Date: 31/12/2024	Generator (Solar)	Bikaner-II PS	ReNew Dinkar Urja Private Limited Solar Power Project – Bikaner-III PS 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity.
4.	0112100017	Tepsol Saur Urja Private Limited	Barmer distt., Rajasthan	20.09.2022	31.03.2025/ 1000	Renewable Power Park Developer (Solar)	Fatehgarh -IV PS	Tepsol Saur Urja Private Limited Renewable Power Park Project – Fatehgarh -IV PS 400 kV S/c (high capacity) line (Suitable to carry minimum 1000 MW per at nominal voltage)

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
								Common Transmission system mentioned at (iii) under ISTS shall also be required for Connectivity.
5.	0112100016	Malur Renewables Pvt. Ltd.	Bikaner distt., Rajasthan	27.09.2022	01.10.2022/ 900 Revised Start Date: 26/06/2025	Renewable Power Park Developer (Solar)	Bhadla-III PS	Malur Renewables Pvt. Ltd. Renewable Power Park Project – Bhadla-III PS 400 kV S/c line (Suitable to carry minimum 900 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
6.	0112100020	Abu Renewables India Private Limited	Jodhpur distt., Rajasthan	30.09.2022	31.12.2025/ 350	Generator (Solar)	Bhadla-III PS	Abu Renewables India Private Limited Solar Power Project – Bhadla-III PS 220 kV S/c line on D/c tower (high capacity) (Suitable to carry minimum 350 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity. <i>St-I & II Connectivity Applications of M/s Abu Renewables were discussed in 12th CMETS meeting, however, Transmission system for St-I Connectivity was not finalized due to GNA Transition, accordingly, St-I Connectivity application was discussed once again in this 14th CMETS- NR meeting. Grant of Intimation shall be issued for both St-I & II Connectivity accordingly.</i>

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
St-I Applications received in the month of October'22								
1.	0112100018	Juniper Green Beta Private Limited	Jodhpur distt., Rajasthan	03.10.2022	31.03.2025/ 400	Generator (Solar)	Bhadla-III PS	Juniper Green Beta Private Limited Solar Power Project – Bhadla-III PS 220 kV S/c line on D/c tower (high capacity) (Suitable to carry minimum 400 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
2.	0112100025	Frugal Energy Private Limited	Jaisalmer distt., Rajasthan	29.10.2022	31.03.2025/ 300	Renewable Power Park Developer (Solar)	Bhadla-III PS	A. For Solar Power Project (50 MW): Through sharing of dedicated transmission system of M/s Juniper Green Beta Private Limited (Appl. No. 0212100027-150 +40 MW) by M/s Frugal Energy –Bhadla-III PS 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage) ➤ Infrastructure required for sharing dedicated transmission system - under the scope of M/s Frugal Energy Private Limited B. For Solar Power Project (250 MW): Frugal Energy Private Limited Renewable Power Park Developer Project – Bhadla-III PS 220 kV S/c (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
3.	0112100026	Luceo Solar Private Limited	Barmer distt., Rajasthan	30.10.2022	31.12.2024/ 200	Generator (Solar)	Fatehgarh -IV PS	Luceo Solar Private Limited Solar Power Project – Fatehgarh -IV PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (iii) under ISTS shall also be required for Connectivity.
4.	0112100027	Bhadla Three Skp Green Ventures Private Limited	Jaisalmer distt., Rajasthan	30.10.2022	31.12.2024/ 300	Renewable Power Park Developer (Solar)	Bhadla-III PS	Bhadla Three Skp Green Ventures Private Limited Renewable Power Park Project – Bhadla-III PS 220 kV S/c line on D/c tower (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
St-I Applications received in the month of November'22								
1.	0112100021	ACME Solar Holdings Private Limited	Jaisalmer distt., Rajasthan	04.11.2022	31.12.2024/ 300	Generator (Solar)	Bhadla-III PS	ACME Solar Holdings Private Limited Solar Power Project – Bhadla-III PS 220 kV D/c line (Suitable to carry minimum 300 MW per circuit at nominal voltage)

Sl. No.	Application ID	Applicant	Project Location	Date of Application	Connectivity Sought Date/(MW)	Nature of Applicant	Location requested for Grant of Stage-I Connectivity	Tr. System for Grant of Stage-I Connectivity
2.	0112100022	ACME Solar Holdings Private Limited	Jaisalmer distt., Rajasthan	04.11.2022	31.12.2024/ 300	Generator (Solar)	Bhadla-III PS	220 kV line to be clubbed with Application Nos. 0112100021 & 0112100022 (300 MW each) of M/s ACME Solar Holdings Private Limited Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
3.	0112100023	ACME Solar Holdings Private Limited	Barmer distt., Rajasthan	04.11.2022	31.03.2025/ 300	Generator (Solar)	Fatehgarh -IV PS	ACME Solar Holdings Private Limited Solar Power Project – Fatehgarh -IV PS 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (iii) under ISTS shall also be required for Connectivity.
4.	0112100028	Sunsure Energy Private Limited	Bikaner distt., Rajasthan	10.11.2022	31.12.2024/ 300	Generator (Solar)	Bikaner-III PS	Sunsure Energy Private Limited Solar Power Project – Bikaner-III PS 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity.

(i) Common Transmission system for Connectivity at Bhadla-III PS (under ISTS)

- Establishment of 1x500 MVA 400/220 kV & 1x1500 MVA 765/400 kV pooling station at Bhadla-III PS
- Bhadla-III– Sikar-II 765 kV D/c line

(ii) Common Transmission system for Connectivity at Bikaner-III PS

- Establishment of 400/220kV, 1x500MVA pooling station at Bikaner-III (765/400/220) kV PS
- LILO of one ckt of 400 kV Bikaner (PG)-Bikaner-II (Quad) D/c line at Bikaner-III PS or Bikaner-II – Bikaner-III 400 kV D/c line (Quad)

(iii) Common Transmission system for Connectivity at Fatehgarh-IV PS (under ISTS)

- Establishment of 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) *
or
- Establishment of 1x1500 MVA, 765/400 kV & 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line(2nd) at Fatehgarh-IV (Section-2) PS

** with minimum capacity of 2100 MVA on each circuit at nominal voltage*

Note: In view of above Common Transmission system required for Connectivity at various ISTS pooling stations, 400/220 kV ICT shall not be required, if Connectivity is being granted at 400 kV. Same shall be suitably included in the respective St-I Intimations.

- It is to mention that the grant of Stage-I Connectivity shall not create any right in favour of the grantee on ISTS infrastructure including bays. Stage-I Connectivity grantee shall be required to update the development of their generation project and associated transmission infrastructure as per format RCON-I-M (available on CTU website>>Stage-I / Stage-II Status Monitoring Portal) by 30th day of June and 31st December of each year.
- Further Stage-I Connectivity grantee is required to secure the Stage-II Connectivity for physical connectivity with the ISTS grid within 24 months. The Stage-I Connectivity grantees who fail to apply for Stage-II Connectivity within 24 months from grant of Stage-I Connectivity shall cease to be Stage-I Grantee and their Application fees shall be forfeited.

3. Stage-II Connectivity applications

It was informed that following Stage-II Connectivity applications were received in the month of Oct'22 & Nov'22 and were agreed to be granted as per the details below:

TABLE 2

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
St-II Applications received in the month of October'22											
1.	0212100027	Juniper Green Beta Private Limited	Jodhpur distt., Rajasthan	03.10.2022	0112100018/400	Generator (Solar)	31.03.2025	150/ 31.03.2025 (Interim) [#]	L&FC	Bhadla-III PS	Common pooling station of M/s Juniper Green Beta Private Limited Solar power project of 150 MW & 40 MW and 50 MW of M/s Frugal – Bhadla-III PS 220 kV S/c line on D/c tower** (Suitable to carry minimum 300 MW at nominal voltage)
2.	0312100013 (Enhancement)	Juniper Green Beta Private Limited	Jodhpur distt., Rajasthan	31.10.2022	0112100018/400	Generator (Solar)	31.03.2025	40/ 31.03.2025 (Interim) [#]	L&FC	Bhadla-III PS	220 kV line to be clubbed with Application Nos. 0212100027, 0312100013 & 0112100025 of M/s Juniper & M/s Frugal Energy Private Limited respectively. - Common 220 kV line bay at Bhadla-III PS shall be developed under ISTS Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
3.	0312100010 (Enhancement)	Juniper Green Stellar Private Limited	Barmer distt., Rajasthan	06.10.2022	1200003672/400	Generator (Solar)	31.01.2025	45/ 31.01.2025 Revised: 30/09/2025 (Interim)##	L&FC	Fatehgarh -IV PS	Common Pooling station of M/s Juniper Green Stellar Private Limited for Solar Power Projects with Appl. Nos. 1200003827(100 MW), 1200003910 (100 MW), 1200003958 (60 MW), 0312100010 (45 MW) & 0312100012 (60 MW) – Fatehgarh-IV PS 220 kV S/c (high capacity) line on D/c tower (Suitable to carry minimum 365 MW at nominal voltage)
4.	0312100012 (Enhancement)	Juniper Green Stellar Private Limited	Barmer distt., Rajasthan	20.10.2022	1200003672/400	Generator (Solar)	31.01.2025	60/ 31.01.2025 Revised: 30/09/2025 (Interim)##	L&FC	Fatehgarh -IV PS	220 kV line to be clubbed with Application Nos. 1200003827, 1200003910, 1200003958, 0312100010 & 0312100012 of M/s Juniper Green Stellar Private Limited 220 kV Common Bay at Fatehgarh-IV PS shall be implemented under ISTS Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity. Note: It was informed that above DTL was earlier proposed to be shared by M/s Cannice Renewables

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
											(Appl.No.-0212100004), however, considering the location of Cannice Generation plant (80 MW) and total St-II Connectivity applications of M/s Juniper (365 MW) at Fatehgarh-IV PS, for better coordination and optimal utilization of DTL & ISTS 220 kV bay, St-II Connectivity to M/s Cannice Renewables was proposed to be revised. Further, M/s Cannice informed during the meeting that they have also applied for St-II enhancement application for 150 MW at Fatehgarh-IV PS. Looking into this development, CTU suggested that grant of St-II Connectivity for these two applications for 80 MW & 150 MW each may be discussed in the next CMETS meeting of NR, wherein separate 220 kV bay grant may be deliberated. M/s Cannice agreed for the same.
5.	0312100011 (Enhancement)	Sprng Nirjara Energy Private Limited	Bikaner distt., Rajasthan	10.10.2022	1200003622/ 400	Generator (Solar)	31.10.2025	50/ 31.10.2025 (Interim)###	L&FC	Bikaner-II PS	Through sharing of dedicated transmission system of M/s TP Saurya Limited (Appl. No. 0212100025-300 MW) by M/s Sprng Nirjara – Bikaner-III PS 220 kV S/c (high capacity) line on D/c tower

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
											(Suitable to carry minimum 350 MW at nominal voltage) Infrastructure required for sharing dedicated transmission system - under the scope of M/s Sprng Nirjara Energy Private Limited 220 kV Common Bay at Bikaner-III PS shall be implemented under ISTS Common Transmission system mentioned at (iii) under ISTS shall also be required for Connectivity.
6.	0212100028	Seven Renewable Power Private Limited	Bikaner distt., Rajasthan@	17.10.2022	0112100011/ 900	Generator (Solar)	30.06.2024	300/ 30.06.2024 Revised by the applicant: 30/01/2025 (Interim) [#]	L&FC	Bhadla-III PS	Seven Renewable Power Private Limited Solar Power Project – Bhadla-III PS 220 kV S/c line (Suitable to carry minimum 300 MW at nominal voltage) Applicant confirmed to build line on S/c tower & 220 kV Common Bay at Fatehgarh-IV PS to be implemented under ISTS. Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
7.	0212100030	Tepsol Sun Sparkle Private Limited	Jaisalmer distt., Rajasthan	25.10.2022	0112100014/ 1000	Renewable Power Park Developer (Solar)	31.03.2025	300/ 31.03.2025 (Interim) [#]	L&A	Bhadla-III PS	Tepsol Sun Sparkle Private Limited Renewable Power Park Project – Bhadla-III PS 220 kV S/c line on D/c tower** (Suitable to carry minimum 300 MW at nominal voltage) 220 kV line bay at Bhadla-III PS shall be developed by the applicant Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.
8.	0212100031	Frugal Energy Private Limited	Jaisalmer distt., Rajasthan	29.10.2022	0112100025/ 300	Renewable Power Park Developer (Solar)	31.03.2025	50/ 31.03.2025 (Interim) [#]	L&A	Bhadla-III PS	Through sharing of dedicated transmission system of M/s Juniper Green Beta Private Limited (Appl. No. 0212100027-150 +40 MW) by M/s Frugal Energy – Bhadla-III PS 220 kV S/c line on D/c tower** (Suitable to carry minimum 300 MW at nominal voltage) Infrastructure required for sharing dedicated transmission system - under the scope of M/s Frugal Energy Private Limited 220 kV line bay at Bhadla-III PS shall be developed under ISTS Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity.

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
St-II Applications received in the month of November'22											
1.	0212100032	Luceo Solar Private Limited	Barmer distt., Rajasthan	07.11.2022	0112100026/ 200	Generator (Solar)	31.12.2024	200/ 31.12.2024 Revised: 30/09/2025 (Interim) ^{##}	L&FC	Fatehgarh -IV PS	Luceo Solar Private Limited Solar Power Project – Fatehgarh -IV PS 220 kV S/c line on D/c tower** 220 kV Bay at Fatehgarh-IV PS shall be implemented under ISTS (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (ii) under ISTS shall also be required for Connectivity.
2.	0212100033	Bhadla Three SKP Green Ventures Private Limited	Jaisalmer distt., Rajasthan	09.11.2022	0112100027/ 300	Generator (Solar)	31/12/2024 ^{\$} (Revised by applicant)	300/ 31.12.2024 ^{\$} (Interim) [#]	L&A	Bhadla-III PS	Bhadla Three SKP Green Ventures Private Limited Solar Power Project – Bhadla-III PS 220 kV S/c line on D/c tower** 220 kV Bay at Bhadla-III PS shall be implemented by the applicant (Suitable to carry minimum 300 MW at nominal voltage) Common Transmission system mentioned at (i) under ISTS shall also be required for Connectivity

Sl. No.	Application No.	Applicant	Location	Date of Application	Application/ Quantum of Stage-I (MW)	Nature of Applicant	Generation Schedule	Stage-II Connectivity Sought (MW)/date	Quantum won / Land & Auditor Basis	Proposed Location as per St-I Application	Tr. System for Grant of Stage-II Connectivity
Earlier granted St-II Applications											
1.	0212100004	Cannice Renewables Energy Private Limited	Barmer distt., Rajasthan	27.08.2022	0112100009/ 500	Generator (Solar)	31.01.2025	80/ 31.01.2025 Revised: 30/09/2025 (Interim)##	L&FC	Fatehgarh -IV PS	It was informed that earlier St-II Connectivity (80 MW) of M/s Cannice was proposed to be shared by M/s Juniper at Fatehgarh-IV PS. Considering the location of Cannice Generation plant (80 MW) and total St-II Connectivity applications of M/s Juniper (365 MW) at Fatehgarh-IV PS, for better coordination and optimal utilization of DTL & ISTS 220 kV bay, St-II Connectivity to M/s Cannice Renewables was proposed to be revised. Further, M/s Cannice informed during the meeting that they have also applied for St-II enhancement application for 150 MW at Fatehgarh-IV PS. Looking into this development, CTU suggested that grant of Connectivity for these two applications for 80 MW & 150 MW each shall be discussed in the next CMETS meeting of NR, wherein separate 220 kV bay may be granted to M/s Cannice and sharing with M/s Juniper may not be required. M/s Cannice agreed for the same.

Tentative Start date of Connectivity at Bhadla-III PS is Dec'24. Final Start date shall be given after award of ISTS work. CTU informed that they are in receipt of number of St-II connectivity applications at Bhadla-III PS. It was opined that establishment of 765/400/220kV Bhadla-III PS along with onwards Corridors i.e. 765 kV Bhadla-III-Sikar-II-Khetri/Narela D/c is under bidding (Ph-III) with 18 months implementation schedule whereas 400kV Bhadla-III-Bhadla(HVDC) 2xD/c & 6 GW Bhadla (HVDC)-Fatehpur HVDC system is approved by NCT (with 42 months schedule) and to be approved by MOP. Considering bidding as well as execution timeline, HVDC system is expected by Dec'26 tentatively. 765/400/220kV Ramgarh PS is also being connected with Bhadla-III PS. Therefore, corridors beyond Bhadla-III PS shall be utilized for transfer of power from RE Generators at Ramgarh/Bhadla-III. As per the above planned system, about 2900 MW can be evacuated utilizing EHVAC corridors i.e. 765kV Bhadla-III-Sikar-II D/c corridor. Out of above, based on receipt of applications, 2600 MW LTA (with connectivity at Ramgarh PS) is already granted utilizing above margins. Therefore, balance about 300 MW margin is available for power transfer beyond Bhadla-III PS utilizing EHV AC corridor, as per the present Ph-III studies. For Power transfer requirement beyond 2900 MW, as per the above planned system, HVDC system shall be utilized for power transfer from Bhadla-III PS onwards

Tentative Start date of Connectivity at Fatehgarh-IV PS (section-2) is Sep'25. Final Start date shall be given after award of ISTS work. For LTA beyond 2000 MW at Fatehgarh-IV PS, Rajasthan SEZ Ph-IV system shall be required, which is being taken up in NCT meeting for approval & schedule for the same may be considered tentatively as Sep'25.

Bikaner-III PS has been approved by NCT & MoP approval is awaited and subsequently tendering shall take place. Accordingly, the tentative Start date of Connectivity is Bikaner-III PS is May'25. Final Start date shall be given after award of ISTS work

** Applicant confirmed the line configuration as S/c line on D/c tower and submitted undertaking for not claiming any overriding priority for ISTS Connectivity at ISTS pooling station through 2nd circuit of the above line. In addition to this, the applicant shall not claim any right over bay allocation at ISTS pooling station on above ground and all issues related to sharing of D/c or M/c tower has to be coordinated among developers themselves.

@ Applicant confirmed the location of plant as Bikaner, distt.

\$ Applicant revised the commissioning schedule of the generation plant as 31/12/2024 vide their email dated 11/01/22

(i) Common Transmission system for Connectivity at Bhadla-III PS (under ISTS)

- Establishment of 1x500 MVA 400/220 kV & 1x1500 MVA 765/400 kV pooling station at Bhadla-III PS
- Bhadla-III– Sikar-II 765 kV D/c line

(ii) Common Transmission system for Connectivity at Fatehgarh-IV PS (under ISTS)

- Establishment 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) *
- or
- Establishment of 1x1500 MVA, 765/400 kV & 1x500 MVA, 400/220 kV Fatehgarh- IV (Section-2) Pooling Station
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line(2nd) at Fatehgarh-IV (Section-2) PS

* with minimum capacity of 2100 MVA on each circuit at nominal voltage

(iii) Common Transmission system for Connectivity at Bikaner-III PS

- Establishment of 400/220kV, 1x500MVA pooling station at Bikaner-III 765/400/220kV PS
- LILO of one ckt of 400 kV Bikaner (PG)-Bikaner-II (Quad) D/c line at Bikaner-III PS or Bikaner-II – Bikaner-III 400 kV D/c line (Quad)

Note: In view of above Common Transmission system required for Connectivity at various ISTS pooling stations, 400/220 kV ICT shall not be required, if Connectivity is being granted at 400 kV. Same shall be suitably included in the respective St-II Intimations.

4. Proposal for grant of LTA to the application received from RE Energy Sources

It was informed that the following LTA applications were received in the month of Oct'22. Details are as below:

TABLE 3

Sl. No.	Application No.	Applicant	LTA Application date (online)	Connectivity Injection Point	Connectivity Application	Drawl Point	Quantum of LTA (MW)	Start Date of LTA	End date of LTA
1	0412100007	XL Xergi Power Private Limited	06.10.2022	Fatehgarh-III PS, Rajasthan	1200002847	GRIDCO (ER)	200	18/01/2024 ^{##}	17/01/2049
2	0412100008	Juniper Green Cosmic Private Limited	29.10.2022	Bikaner-II PS, Rajasthan	1200003740	Target (NR)	100 (Target)	31/01/2025	31/01/2050

^{##}Interim Start date of LTA shall be 31/12/24 as per the deliberations below:

For application at Sl. No. 1: It was informed that Stage-II Connectivity (1200002847, 400 MW) was granted to M/s XL Xergi Power Private Limited at Fatehgarh-III PS through 220 kV S/c line. 1 No. 220 kV bay at Fatehgarh-III PS is being implemented under ISTS. Applicant confirmed the same. Now, M/s XL Xergi has applied for LTA of 200 MW on firm basis to GRIDCO-ER (200MW).

It was deliberated that based on the system studies, the required power transfer of 200 MW from Fatehgarh-III PS is envisaged with the Transmission system associated with SEZ in Rajasthan under 20 GW Phase-III scheme. The transmission system has been approved in 3rd NRPC(TP) meeting held on 19/02/2021 & 5th NCT meeting held on 25/08/2021 & 02/09/2021. It was informed that BPCs had approached Hon'ble Supreme Court committee for GIB clearance for Ph-III packages, out of which, few packages are already accorded

with GIB committee ratification. However, other packages are also linked to the packages which are granted ratification by GIB Supreme Committee, which are yet awaiting the committee clearance.

Therefore, considering the above status and scheduled bid submission in Jan'23 for some packages, the tentative timeline for the above scheme can be considered as Dec'24. Same shall be confirmed based on GIB Supreme Court committee clearance for other balance packages as well as award of the system. Applicant noted the same.

In addition, North-West Inter-regional system strengthening scheme shall also be required for LTA:

Phase-I (For Short term) [Commissioning schedule: May'24.]

1. Bypassing of 400 kV Kankroli - Bhinmal-Zerda line at Bhinmal to form 400 kV Kankroli – Zerda (direct) line #
2. Reconductoring of 400 kV Jodhpur (Surpura)(RVPN) – Kankroli S/c (twin moose) line with twin HTLS conductor*-188 km

with necessary arrangement for bypassing Kankroli- Zerda line at Bhinmal with suitable switching equipment inside the Bhinmal substation.

* with minimum capacity of 1940 MVA/ckt at nominal voltage; Upgradation of existing 400kV bay equipment's each at Jodhpur (Surpura)(RVPN) and Kankroli S/s (3150 A)

It was noted that the scheme has been awarded with tentative timelines as Nov'23 for Bypassing arrangement & May'24 for Reconductoring part. Details of transmission system for present LTA is mentioned at **Annexure-I**.

It was noted that the LTA has been applied on firm basis, and for further drawl of power by ER discoms through STU network from ISTS, NOC from the concerned STU shall be required. Accordingly, applicant was requested to take up the matter of NoC with GRIDCO (ER). Subsequently, applicant vide their email dated 29/12/2022 submitted the NoC issued by OPTCL.

Accordingly, it was agreed to grant LTA to M/s XI Xergi Power Private Limited for 200 MW from Fatehgarh-III PS to GRIDCO- ER (200 MW) from 31/12/2024 (Interim) to 17/01/2049.

For application at Sl. No. 2: It was informed that Stage-II Connectivity (1200003740, 100 MW) was granted to M/s Juniper Green Cosmic Private Limited at Bikaner-II PS through 220 kV S/c line through sharing of DTS of M/s Sprng Nirjara Energy Private Limited. 1 No. 220 kV bay at Bikaner-II PS is being implemented by M/s Sprng, which is common for both the RE developers. Applicant confirmed the same. Now, M/s Juniper has applied for LTA of 100 MW on target basis to NR (100MW).

It was deliberated that based on the system studies, Rajasthan SEZ Phase-I & II Transmission schemes is already agreed in 2nd and 5th NRSCT meetings held on 13/11/2018 & 13/09/2019 respectively. The required power transfer of 100 MW from Bikaner-II PS was

envisaged under the said system. In addition to this, the transmission system for present LTA, consists of Part-G of Ph-II Tr. System with commissioning schedule of Nov'23.

In addition, North-West Inter-regional system strengthening scheme shall also be required for LTA:

Phase-I (For Short term) [Commissioning schedule: May'24.]

1. Bypassing of 400 kV Kankroli - Bhinmal-Zerda line at Bhinmal to form 400 kV Kankroli – Zerda (direct) line #
2. Reconductoring of 400 kV Jodhpur (Surpura)(RVPN) – Kankroli S/c (twin moose) line with twin HTLS conductor*-188 km

with necessary arrangement for bypassing Kankroli- Zerda line at Bhinmal with suitable switching equipment inside the Bhinmal substation.

* with minimum capacity of 1940 MVA/ckt at nominal voltage; Upgradation of existing 400kV bay equipment's each at Jodhpur (Surpura)(RVPN) and Kankroli S/s (3150 A)

Above scheme has been awarded with tentative timelines as Nov'23 for Bypassing arrangement & May'24 for Reconductoring part. Details of transmission system for present LTA is mentioned at **Annexure-II**.

Accordingly, it was agreed to grant LTA to M/s Juniper Green Cosmic Private Limited for 100 MW from Bikaner-II PS to Target NR (100 MW) from 31/01/2025 to 31/01/2050.

5. Conventional Connectivity Application from M/s Hindustan Zinc Limited for 200 MW as Bulk Consumer

It was deliberated that following 1 No. of Conventional Connectivity application from M/s Hindustan Zinc Limited was received in the month of July'22 with details as under:

Sl. No.	Application ID	Name of the Applicant	Application Type	Submission Date	Region	Project Location	Start Date of Connectivity	Quantum (MW)
1	0012100007	Hindustan Zinc Limited	Bulk Consumer	25.07.2022	NR	Rajasamand distt., Rajasthan	01.07.2024	200

- It was informed that Connectivity application from M/s HZL was deliberated in the 10th, 11th & 12th CMETS-NR meeting held on 30/08/2022, 30/09/2022 & 28/10/2022 respectively. It was informed that M/s Hindustan Zinc Ltd. (HZL) had applied for connectivity to the ISTS for 200 MW as a bulk consumer located at Rajasamand distt., Rajasthan with start date of Connectivity as 01/07/2024. As per the application, the bulk consumer entity is located close to the existing 400/220 kV Kankroli (PG) substation. Accordingly,

following transmission system was proposed for Connectivity to M/s HZL for 200 MW as Bulk Consumer in Rajasamand distt., Rajasthan:

- HZL– Kankroli (PG) 220kV D/c line along with associated line bays at both end- Cost about 50 Cr
- In the previous meetings, it was deliberated that as clause 1.4 of CERC Detailed Procedure *“The applicant (Generator/bulk consumer) already connected to grid (regional or state grid) or for which connectivity is already granted under the present arrangement, shall not be allowed to apply for additional connectivity for the same capacity. In case of extension of capacity of generator or bulk consumer, however, it shall be required to make application for connectivity as per the provisions of these procedures.”* Further, it was noted that in the CERC regulation, there is no provision of Dual Connectivity for same load as mentioned above, therefore, in case HZL is getting its demand served from RVPN/Rajasthan Discoms, have HZL approached/informed the utility for disconnection of connected load. HZL earlier informed that they have total load of 193 MW for which they have 2x91 MW captive thermal generation as well as 95 MVA sanctioned load from RVPN/DISCOM.
- Accordingly, CTU earlier stated that as there is no provision of Dual Connectivity for same load as mentioned above, therefore, as RVPNL has sanctioned load of 95 MW to HZL, following was proposed:
 - HZL can apply for remaining quantum of 105 MW for Connectivity in ISTS. As current application of 200 MW quantum shall have change of more than 40% qualifying as material change in quantum of power to be interchanged necessitating a fresh application for 105 MW as per CERC connectivity regulation.
 - For disconnection of this 95 MW from STU/DISCOM network, HZL may take up the matter with RVPNL and present connectivity application can be discussed in the subsequent CMETS-NR meeting after final deliberations between RVPNL & HZL. HZL will inform on such development at the earliest so that further necessary action regarding application can be taken.
 - Accordingly, based on above, it was decided that HZL shall revert on their current application of 200 MW as bulk consumer in ISTS.
- However, HZL/RVPN response was awaited in this regard. Subsequently, RVPNL submitted NoC to providing Connectivity vide letter dated 22/12/22 to HZL at Kankroli Substation (PG). However, it was deliberated that M/s HZL shall abide with all the terms & conditions mentioned in the NoC issued by RVPNL.
- Further, it was also informed that there is no provision in the Electricity Act or in the 2009 Connectivity Regulations for construction of a dedicated transmission line by a bulk consumer. The same has also been noted by the Hon’ble Commission in the orders given to various Bulk consumers and has stated that “We are not inclined to issue any general directions in this regard and such matters shall be decided on case to case basis depending upon specific circumstances and facts of the case”.

- Further, CTU also stated that for 200 MW power transfer, based on receipt of LTA, requirement of 400/220kV ICT at Kankroli shall be studied.
- Accordingly, it was advised that M/s HZL shall have to approach CERC for decision on implementation modality of the dedicated line in instant case. Therefore, the transmission system as stated above was agreed for grant of Connectivity to M/s HZL for 200 MW as Bulk Consumer:
 - HZL– Kankroli (PG) 220kV D/c line along with associated line bays at both end (under the scope of applicant) [Line capacity shall be suitable to carry minimum 300 MW per circuit at nominal voltage]

B. ISTS Expansion

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :7.5GW) (Jaisalmer/Barmer Complex)

It was deliberated that Renewable Energy Zones (REZs) were identified by MNRE/SECI with a total capacity of 181.5 GW for likely benefits by the year 2030 in eight states, which includes 75 GW REZ potential in Rajasthan comprising of 15 GW Wind and 60 GW Solar. In this regard a Committee on Transmission Planning for RE was constituted by MOP for planning of the requisite Inter State Transmission System required for the targeted RE capacity by 2030.

In this regard, a Comprehensive transmission plan for evacuation of 75GW RE potential from Rajasthan is evolved. As part of committee report as well as MNRE/SECI inputs, 12GW potential (Wind: 6GW, Solar: 6GW) along with 4GW BESS was envisaged at Fatehgarh-IV in Jaisalmer complex by 2030. Out of 12GW potential, 8GW potential (Wind: 4GW, Solar: 4GW) along with 2GW BESS considered by 2027 at Fatehgarh-IV PS.

Similarly, 7GW potential (Wind: 3GW, Solar: 4GW) along with 1.5 GW BESS envisaged at Barmer-I in Barmer complex by 2030. Out of 7GW potential, 6GW potential (Wind: 3GW, Solar: 3GW) along with 1GW BESS considered in Barmer-I by 2022.

S.No	Pooling Station	Total RE Potential (by 2030)		RE Potential (by 2027)		Net (Max) Dispatch Considered by 2027(GW)	Application Status
		Source	Capacity (GW)	Capacity (GW)			
1	Fatehgarh-IV	Wind	6	4	4		

		Solar	6	4		At present St-II Connectivity for 3.1 GW RE, against the potential of 2.1 GW (under Ph-III), is already received at Fatehgarh-IV (Section-1).
		BESS	4	2		
2	Barmer-I	Wind	3	3	3.5	No application received yet
		Solar	4	3		
		BESS	1.5	1		
		Total (GW)			7.5	

Solar Dispatch considered @100%, Wind dispatch considered @50% in June Solar Max Scenario

It was stated that as part of Rajasthan REZ Ph-III (20GW) transmission system, 400/220kV Fatehgarh-IV PS and onward transmission system is already under bidding. The evacuation system planned earlier in Ph-III from 400/220kV Fatehgarh-IV PS is adequate for evacuation of about 2 GW RE capacity. Considering above, space provision for separate 765/400//220kV (Section-2) at Fatehgarh-IV PS was approved in 8th NCT meeting held on 25.03.22

At present, St-II Connectivity for about 3.1 GW RE is already received at Fatehgarh-IV PS which has exceeded the envisaged potential of Fatehgarh-IV PS [Ph-III potential: 2.1 GW). Considering grant of connectivity to new RE generators at Fatehgarh-IV PS as well as for evacuation of power from Fatehgarh-IV PS beyond 2 GW in Jaisalmer Complex and Barmer-I PS in Barmer complex, 765kV high capacity corridors will be required from above pooling stations.

Accordingly, system studies were carried out and study plots along with network files were sent to all stakeholders dated 30.11.22. A Joint Study Meeting was held in virtual mode on 05.12.22 with NRPC, CEA, GRID-INDIA (Formerly known as POSOCO), RVPN, HVPN, UPPTCL, PSTCL and other STUs of Northern region to finalize the Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2 :7.5GW) (Jaisalmer/Barmer Complex).

Based on observations received from Stakeholders as well as discussion held in joint meeting, revised study files were shared along with evening peak/off peak files to all constituents on 16.12.22 incorporating the suggestions and inputs from UP, Punjab & GRID-INDIA/NRLDC. In Feb solar maximized scenario (revised case) loading of 400kV RAPP- Shujalpur D/c line was marginally higher (about 910MW) in N-1 contingency. The loading of above line will be reviewed with progress of RE generation projects at Rajasthan

and strengthening requirement will be identified later, if required. In revised and additional cases, line loadings of other lines were found to be in order as well as voltages were within limits on all substations. In the meeting, CTU enquired to RVPN about drawl requirement at Jalore PS. RVPN replied that they have not envisaged any drawl requirement in Jalore area. Accordingly, scope for RVPN drawl requirement from future provision at Jalore PS was deleted in final scope.

The proposed comprehensive Transmission scheme as agreed in Joint study meeting and present CMETS-NR meeting is as under:

Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2: 7.5 GW)

Fatehgarh-IV PS: 4GW (4GW Solar, 4GW Wind, 2GW BESS) and Barmer-I PS: 3.5GW (3 GW Solar, 3 GW Wind, 1 GW BESS)

Northern Region

A. Fatehgarh-IV PS: 4 GW (4GW (Solar) + 4GW (Wind) + 2 GW BESS)

- Establishment of 4x1500 MVA, 765/400 kV & 5x500 MVA[^], 400/220 kV Fatehgarh- IV (Section-2) Pooling Station along with 2x240 MVA (765 kV) Bus Reactor & 2x125 MVA (400 kV) Bus Reactor *
- Fatehgarh-IV (Section-2) PS – Bhinmal (PG) 400 kV D/c line (Twin HTLS) along with 50 MVA switchable line reactor on each ckt at each end (~200 km)
- LILO of both ckts of 765 kV Fatehgarh-III- Beawar D/c line (2nd) at Fatehgarh-IV (Section-2) PS along with 330 MVA switchable line reactors at Fatehgarh-IV PS end of each ckt of 765 kV Fatehgarh-IV PS - Beawar D/c line** (formed after LILO) (~15 km)
- Beawar- Mandsaur PS 765 kV D/c line along with 240 MVA switchable line reactor on each circuit at each end (~250 km)
- Augmentation by 1x1500 MVA, 765/400 kV ICT at Fatehgarh-II PS (7th)
- 6 nos. of 220 kV line bays at Fatehgarh-IV PS (for RE connectivity)
- 220kV Sectionalization bay (1 set) along with BC (2 nos.) and TBC (2 nos.) at Fatehgarh- IV PS

Future provisions at Fatehgarh-IV PS is already approved in 8th NCT meeting dated 25.03.22

[^]incl 1x500MVA ICT to fulfill 'N-1' requirement

**Incl. spare one ICT unit (500MVA) and spare one reactor unit (80MVA) at Fatehgarh-IV PS*

***Incl spare one reactor unit (110MVA) at Fatehgarh-IV PS*

B. Barmer-I PS: 3.5GW (3GW (Solar) + 3GW (Wind) +1 GW BESS)

- Establishment of 4x1500 MVA, 765/400 kV & 5x500 MVA[^], 400/220 kV Barmer-I Pooling Station along with 2x240 MVA (765 kV) Bus Reactor & 2x125 MVA (400 kV) Bus Reactor*
- Fatehgarh-III (Section-2) PS – Barmer-I PS 400 kV D/c line (Quad) (~50 km)
- 6 nos. of 220 kV line bays at Barmer-I PS (for RE connectivity)

- 220kV Sectionalization bay (1 set) along with BC (2 nos.) and TBC (2 nos.) at Barmer-I PS

Future provisions at Barmer-I PS:

Space for

- 765/400kV ICT along with bays- 2 no.
- 765 kV line bays along with switchable line reactors – 4 nos.
- 765kV Bus Reactor along with bay: 1 no.
- 400 kV line bays –4 no.
- 400 kV line bays along with switchable line reactor –4 nos.
- 400/220kV ICT along with bays -5 nos.
- 400 kV Bus Reactor along with bay: 1 no.
- 400kV Sectionalization bays: 2 sets
- 220 kV line bays for connectivity of RE Applications - 8 nos.
- 220kV Sectionalization bay: 2 sets
- BC (2 nos.) & TBC (2 nos.)
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar)

[^]incl 1x500MVA ICT to fulfill 'N-1' requirement

**Incl. spare one ICT unit (500MVA) and spare one reactor unit (80MVar) at Barmer-I PS*

C. Common Transmission System for RE projects at Fatehgarh-IV and Barmer-I in NR

- Establishment of 2x1500 MVA, 765/400 kV Substation at suitable location near Jalore along with 2x240 MVar (765 kV) & 2x125 MVar (400 kV) Bus Reactor*
- Fatehgarh-IV (Section-2) PS–Jalore PS 765 kV D/c line along with 240 MVar switchable line reactor for each circuit at each end (~200 km)
- Barmer-I PS– Jalore PS 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at Jalore PS end** (~165 km)
- Jalore PS-Chittorgarh (PG) 400 kV D/c line along with 50 MVar switchable line reactor for each circuit at each end (Quad) (~200 km)
- Jalore PS- Mandsaur PS 765 kV D/c line along with 330 MVar switchable line reactor for each circuit at each end (~320 km)

Future provisions at Jalore PS:

Space for

- 765/400kV ICT along with bays- 4
- 765 kV line bays along with switchable line reactors – 4

- 765kV Bus Reactor along with bay: 1 nos.
- 400 kV line bays along with switchable line reactor –4
- 400 kV line bays –4
- 400 kV Bus Reactor along with bay: 1 no.
- 400kV Sectionalization bay: 2 sets
- 400/220kV ICT along with bay - 6 no.
- 220kV line bays -10
- 220kV Sectionalization bay: 2 sets
- BC (3 nos.) & TBC (3 nos.)
- STATCOM (2x±300MVar) along with MSC (4x125 MVar) & MSR (2x125 MVar)

**Incl. spare one ICT unit (500MVA) and spare one reactor unit (80MVar) at Jalore PS*

***Incl spare one reactor unit (110MVar) at Jalore PS end*

Estimated Cost : Rs 13,656.05 Cr (NR Portion)

Implementation Time-frame: 24 months

In addition to above, transmission scheme is also required in Western region for further dispersal of power for above RE complexes (Fatehgarh-IV and Barmer-I). The transmission system for WR portion was discussed in a meeting amongst CEA, CTUIL, MPPTCL and GRID-INDIA on 01.12.2022, wherein the system was in-principally agreed. The scheme was further agreed in 13th CMETS-WR meeting held on 07.12.2022, which is as under

Western Region

Common Transmission System for Fatehgarh-IV (4GW) and Barmer-I (3.5GW) in Western Region

- Establishment of 765kV Mandsaur Pooling Station along with 2x330MVar (765kV) Bus Reactors (with 1x110MVar & 1x80MVar, 765kV spare single phase reactor unit for line/bus reactor)

Future Scope (Space For):

- 765/400kV ICT along with bays- 6 no.
- 765 kV line bays along with switchable line reactors – 6 nos.
- 765kV Bus Reactor along with bay: 2 no.
- 765kV Sectionalizer bay: 1 -set
- 400 kV line bays along with switchable line reactor – 12 nos.

- 400/220kV ICT along with bays -8 nos.
 - 400 kV Bus Reactor along with bay: 2 no.
 - 400kV Sectionalization bay: 1- set
 - 220 kV line bays: 16 nos.
 - 220kV Sectionalization bay: 2 sets
 - 220kV BC and TBC: 3 nos.
 - STATCOM (± 300 MVAr) along with MSC (2x125 MVAr) & MSR (1x125 MVAr): 1 no.
- Mandsaur PS – Indore(PG) 765 kV D/c Line (200km) along with 1x330MVAr switchable line reactor (SLR) on each ckt at Mandsaur end
 - Establishment of 765/400 (2x1500MVA) 400/220 (2x500MVA) & 220/132kV (3x200MVA) Kurawar S/s (with 1x500MVA spare single phase transformer unit) with 2x330MVAr 765kV bus reactor and 1x125MVAr 420kV bus reactor (with 1x110MVAr & 1x80MVAr, 765kV spare single phase reactor unit for line/bus reactor)

Future Scope:

Space for

- 765/400kV ICT along with bays- 4 no.
 - 765 kV line bays along with switchable line reactors – 8 nos.
 - 765kV Bus Reactor along with bay: 2 no.
 - 765kV Sectionalizer bay: 1 -set
 - 400 kV line bays along with switchable line reactor – 10 nos.
 - 400/220kV ICT along with bays -6 nos.
 - 400 kV Bus Reactor along with bay: 2 no.
 - 400kV Sectionalization bay: 1- set
 - 220 kV line bays: 12 nos.
 - 220kV Sectionalization bay: 2 sets
 - 220kV BC and TBC: 3 nos.
 - 220/132kV ICT along with bays: 3 nos.
 - 132kV line bays: 8
 - 132kV Sectionalization bay: 1 set
 - STATCOM (± 300 MVAr) along with MSC (2x125 MVAr) & MSR (1x125 MVAr): 1 no.
- Mandsaur – Kurawar 765kV D/c line (~235km.) with 240MVAr switchable line reactors at both ends
 - LILO of Indore – Bhopal 765kV S/c line at Kurawar (LILO length ~15km.)

- Kurawar – Ashtha 400kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line (~65km.)
- LILO of one circuit of Indore – Itarsi 400kV D/c line at Astha (LILO length ~ 30km.)
- Shujalpur – Kurawar 400kV D/c (Quad ACSR/AAAC/AL59 moose equivalent) line (~40km.)

Implementation Time-frame: 24 months

Estimated Cost : Rs 5967.12 Cr (WR Portion)

The transmission scheme Map for proposed scheme is as under

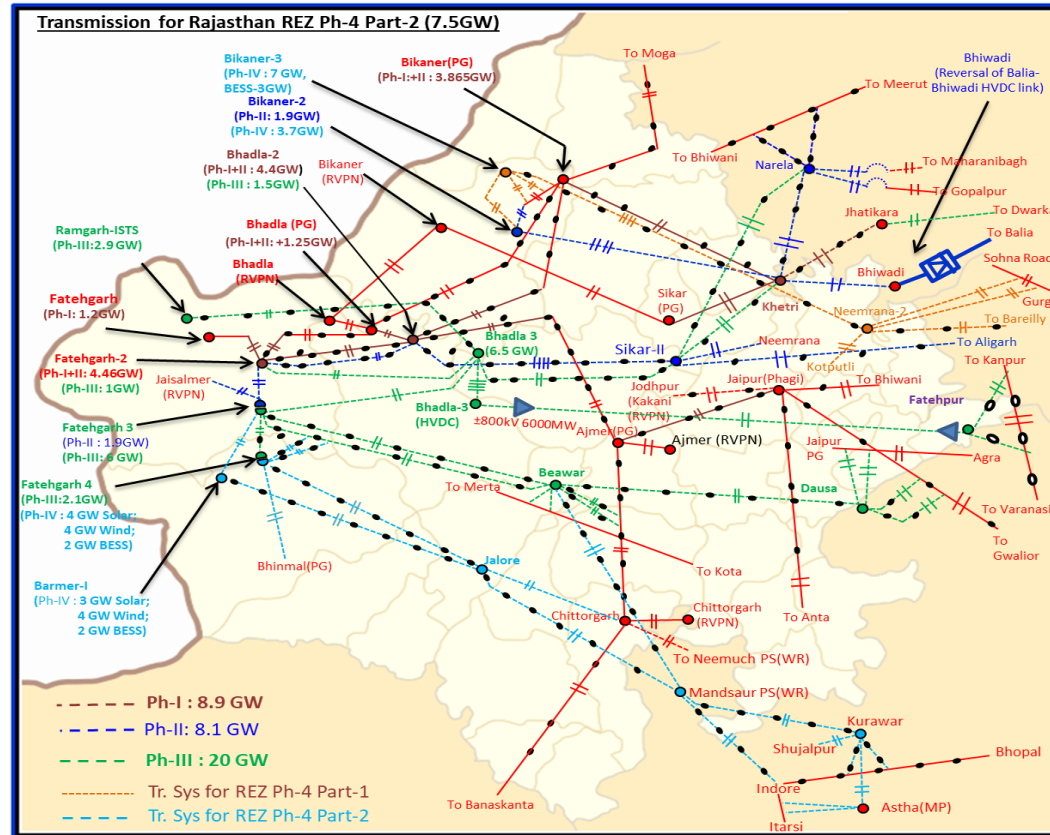


Fig 1: Transmission system for evacuation of power from Rajasthan REZ Ph-IV (Part-2) (Jaisalmer/Barmer Complex)

--X--X--X--

The following provisions shall be applied with respect to grant of connectivity to **RE projects** to the ISTS Grid:

- i. The grant of Stage-I Connectivity shall not create any vested right in favor of the grantee on ISTS infrastructure including bays. Stage-I Connectivity grantee shall be required to update the quarterly progress of development of their generation project and associated transmission infrastructure/dedicated line as per format **RCON-I-M** by 30th day of June and 31st day of December of each year. Further, Stage-I Connectivity grantees who fail to apply for Stage-II Connectivity within 24 months from grant of Stage-I Connectivity shall cease to be Stage-I grantee and their Application fees shall be forfeited.
- ii. If the capacity of the Stage-I Connectivity location is allocated to other Stage-II grantees, the balance Stage-I grantees shall be allocated Stage-II Connectivity to an alternate location.
- iii. For optimization of ROW, it is proposed that transmission towers of various dedicated lines, up to 2-3 km periphery from the entry of the pooling station may be of **Multi-circuit type**. The generation developers/applicants associated with the various pooling stations may coordinate amongst themselves for implementation of such span/stretch of the dedicated connectivity lines with M/C towers.
- iv. The grant of Stage-I / Stage-II Connectivity shall not entitle an applicant to interchange any power with the grid unless it obtains LTA/MTOA/STOA for power transfer requirements. As grant of LTA may require transmission system strengthening, the applicants are advised to apply for LTA immediately, to enable timely transfer of power to the beneficiaries. Also as per the prevailing Transmission Planning Criteria of CEA, "N-1" contingency criteria may not be applied to the immediate connectivity of wind/solar farms with the ISTS/Intra State grid.
- v. Stage-II Connectivity Grantee shall sign the Transmission Agreement for Connectivity and submit the Connectivity Bank Guarantee (**Conn-BG1 and Conn-BG2**) to CTU within 30 days of issue of intimation. No extension of time shall be granted and in case of failure to sign the Agreement and / or to furnish the requisite bank guarantee, Stage-II Connectivity shall be cancelled under intimation to the grantee without any prior notice.
- vi. Two or more Applicants may apply for Stage-II Connectivity at a common bay along with an agreement duly signed between such Applicants for sharing the Dedicated Transmission Line. The Stage-II Connectivity shall be granted to such Applicants subject to availability of capacity in the Dedicated Transmission Line. In such cases, **Conn-BG1 and Conn-BG2**, as applicable as per Clause 10.8 of the Revised RE Procedure, shall be submitted by each such Applicant.
- vii. The Stage-II Connectivity grantee shall furnish progress of the monitoring parameters on quarterly basis in the format given at **FORMAT-RCON-II-M** by the last day of each quarter. Failure to update progress of the monitoring parameters shall be considered as adverse progress and in such case CTU shall approach the Commission for appropriate directions. Further, the

Stage-II Connectivity grantees shall be required to complete the dedicated transmission line(s) and pooling sub-station(s) as defined under para 11.2(A) of the Revised Detailed Procedure for RE. If the grantee fails to complete the dedicated transmission line within the stipulated period, the Conn-BG1 & Conn-BG2 of the grantee shall be encashed and Stage-II connectivity shall be revoked.

- viii. Applicants after grant of Stage-II Connectivity to the grid shall have to furnish additional details to CTU for signing of “Connection Agreement” as per format given at **FORMAT-CON-4**. The finalized template of generation data for Solar and Wind based generation projects has been appended as additional sections at Para F (Details of Connection – Solar PV Station) and Para G (Details of Connection – Wind Generating Station) in the existing FORMAT-CON-4. The modified FORMAT CON-4 is available on our website (www.powergridindia.com >> CTU Open Access). The Applicants are advised to furnish such details as early as possible for enabling them have lead time for any type of access.
- ix. The CTU will process the above information and will intimate the Connection details as per format given at **FORMAT-CON-5**. Pursuant to such Connection details, the applicant shall have to sign “Connection Agreement” with CTU prior to the physical inter-connection as per format given at **FORMAT-CON-6**. In case the connectivity is granted to the ISTS of an inter-State transmission licensee other than the CTU, a tripartite agreement shall be signed between the applicant, the Central Transmission Utility and such inter-State transmission licensee, in line with the provisions of the Regulations.
- x. Applicants are required to submit the test reports supported vide undertaking as well as compliance certificate from manufacturer for all applicable provisions under the CEA (Technical Standards for Connectivity to the Grid) Regulations, 2007 (as amended) (including the provisions of LVRT/HVRT, active power injection control, dynamically varying reactive power support, limits for Harmonic & DC current injection, Flicker limits etc.) from labs accredited by Govt./NABL/other recognized agencies. In case any discrepancies / incompleteness are found in the documents / test reports submitted to CTU, the connection offer (CON-5) / connection agreement (CON-6) shall not be processed further.
- xi. CEA vide order dated 09.11.2020, has mandated all power generating stations of 0.5 MW or above capacity to register themselves on CEA e-portal and get a **Unique Registration Number (URN)**. The same is required as per Regulation 11 of Technical Standards for Connectivity with the Grid Regulations, 2007 prior to physical inter-connection of the Generating station with ISTS Grid.

Transmission system for LTA (200 MW) at Fatehgarh-III PS**A. Transmission system for present LTA (Appl. No. 0412100007) at Fatehgarh-III PS**

- 1) Augmentation of 1x500 MVA (3rd), 400/220kV ICT at Fatehgarh-3 pooling station (Section-II)
- 2) Augmentation of 2x1500 MVA (3rd and 4th), 765/400kV ICT at Fatehgarh-3 pooling station (Section-II)

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under Phase-III (20 GW) scheme)

- 1) Establishment of 2x1500MVA 765/400kV Substation at suitable location near Beawar along with 2x330 MVA_r, 765kV Bus Reactor & 2x125 MVA_r, 420kV bus Reactor
- 2) Fatehgarh-3– Beawar 765 kV D/c (2nd) along with 330 MVA_r Switchable line reactor for each circuit at each end of Fatehgarh-3– Beawar 765 kV D/c line (2nd)
- 3) Fatehgarh 3- Bhadla-3 400kV D/c line (Quad) along with 50 MVA_r Switchable line reactor for each circuit at both ends of Fatehgarh 3- Bhadla-3 400kV D/c line
- 4) LILO of both circuit of Ajmer-Chittorgarh 765 kV D/c line at Beawar substation
- 5) LILO of 400kV Kota –Merta line at Beawar substation
- 6) Establishment of 2x1500 MVA 765/400kV substation at suitable location near Dausa along with 2x330 MVA_r, 765 kV Bus Reactor & 2x125 MVA_r, 420 kV bus Reactor
- 7) LILO of both circuits of Jaipur (Phagi)- Gwalior 765 kV 2xS/c at Dausa along with 240 MVA_r Switchable line reactor for each circuit at Dausa end of Dausa – Gwalior 765 kV D/c line
- 8) LILO of both circuits of Agra – Jaipur(south) 400kV D/c at Dausa along with 50 MVA_r Switchable line reactor for each circuit at Dausa end of Dausa – Agra 400kV D/c line
- 9) Beawar – Dausa 765 kV D/c line along with 240 MVA_r Switchable line reactor for each circuit at each end
- 10) STATCOM:
Fatehgarh – III S/s: STATCOM: $\pm 2 \times 300$ MVA_r, 4x125 MVA_r MSC, 2x125 MVA_r MSR

Additional Scheme to relieve overloading of 400 kV Bhinmal-Zerda line (Twin Moose)

- 1) Bypassing of 400 kV Kankroli - Bhinmal-Zerda line at Bhinmal to form 400 kV Kankroli – Zerda (direct) line #
- 2) Reconductoring of 400 kV Jodhpur (Surpura)(RVPN) – Kankroli S/c (twin moose) line with twin HTLS conductor*-188 km
with necessary arrangement for bypassing Kankroli- Zerda line at Bhinmal with suitable switching equipment inside the Bhinmal substation.* with minimum capacity of 1940 MVA/ckt at nominal voltage; Upgradation of existing 400kV bay equipments each at Jodhpur (Surpura)(RVPN) and Kankroli S/s(3150 A)

Transmission system for LTA (100 MW) at Bikaner-II PS**A. Transmission system for present LTA (Appl. No. 0412100008) at Bikaner -II PS**

- 1) Augmentation of 1x500MVA (2nd) 400/220 kV ICT at Bikaner –II PS

B. Common Transmission system (Part of Transmission system associated with SEZ in Rajasthan under 8.1 GW Phase-II scheme)

- 1) Removal of LILO of one circuit of Bhadla-Bikaner (RVPN) 400kV D/c(Quad) line at Bikaner (PG). Extension of above LILO section from Bikaner (PG) upto Bikaner-II PS to form Bikaner-II PS – Bikaner (PG) 400kV D/c(Quad) line
- 2) Bikaner-II PS – Khetri 400 kV 2xD/c line (Twin HTLS* on M/c Tower)
- 3) 1x80MVA switchable Line reactor on each circuit at Khetri end of Bikaner-II – Khetri 400 kV 2xD/c Line
- 4) Khetri- Bhiwadi 400 kV D/c line (Twin HTLS)*
- 5) Augmentation of 1x1500 MVA (3rd), 765/400kV ICT at Bikaner (PG)
- 6) Establishment of 765/400 kV, 3X1500 MVA GIS substation at Narela with 765 kV (2x330 MVA) bus reactor and 400kV (1x125 MVA) bus reactor.
- 7) Khetri – Narela 765 kV D/c line
- 8) LILO of 765 kV Meerut- Bhiwani S/c line at Narela
- 9) 1x330 MVA Switchable line reactor for each circuit at Narela end of Khetri – Narela 765kV D/c line
- 10) Removal of LILO of Bawana – Mandola 400kV D/c (Quad) line at Maharani Bagh /Gopalpur S/s. Extension of above LILO section from Maharani Bagh/ Gopalpur upto Narela S/s so as to form Maharani Bagh – Narela 400kV D/c(Quad) and Maharani Bagh-Gopalpur-Narela 400 kV D/c (Quad) lines.
- 11) 2 no of line bays at Narela each for Maharani Bagh – Narela 400 kV D/c (Quad) and Maharani Bagh –Gopalpur**-Narela 400 kV D/c (Quad) lines formed after removal of LILO of Bawana – Mandola 400kV D/c(Quad) line at Maharani Bagh/Gopalpur S/s and Extension of above LILO section from Maharani Bagh/Gopalpur upto Narela S/s.
- 12) $\pm$ 300 MVA, STATCOM along with 2x125 MVA MSC, 1x125 MVA MSR at Bikaner-II
- 13) Power reversal on $\pm$ 500 KV, 2500 Balia- Bhiwadi HVDC line upto 2000 MW from Bhiwadi to Balia - Power reversal in Balia-Bhiwadi HVDC line

*with minimum capacity of 2100 MVA on each circuit at nominal voltage

**Gopalpur Substation is under implementation by DTL by LILO of Bawana – Maharani Bagh 400kV D/c(Quad) at Gopalpur substation.

Additional Scheme to relieve overloading of 400 kV Bhinmal-Zerda line (Twin Moose)

1. Bypassing of 400 kV Kankroli - Bhinmal-Zerda line at Bhinmal to form 400 kV Kankroli – Zerda (direct) line #
2. Reconductoring of 400 kV Jodhpur (Surpura)(RVPN) – Kankroli S/c (twin moose) line with twin HTLS conductor*-188 km

with necessary arrangement for bypassing Kankroli- Zerda line at Bhinmal with suitable switching equipment inside the Bhinmal substation.* with minimum capacity of 1940 MVA/ckt at nominal voltage; Upgradation of existing 400kV bay equipments each at Jodhpur (Surpura)(RVPN) and Kankroli S/s(3150 A)

List of Participants of 14th Consultation meeting for Evolving Transmission Schemes in NR held on 23.12.2022

SECI

Shri R. K. Agarwal Consultant

POSOCO

Shri Gaurav Malviya Manager

CTU

Shri Jasbir Singh	CGM (CTU)
Shri Kashish Bhambhani	GM (CTU)
Shri Sandeep Kumawat	Ch. Manager (CTU)
Smt. Ankita Singh	Ch. Manager (CTU)
Shri R Narendra Sathvik	Manager (CTU)
Shri Yatin Sharma	Dy. Manager (CTU)
Shri Roushan Kumar	Engineer (CTU)
Shri Madhusudan Meena	Engineer (CTU)

DTL

Shri Dinesh Singh Manager Planning

RVPN

Shri Sanjay Mathur Executive Engineer

HVPNL

Shri Rajesh Kumar Jangra

Executive Engineer/ System Study

PTCUL

Shri Ravi Shankar

LTA/Connectivity Applicants

Shri Radheshyam Goyal

Tepsol Sun Sparkle Pvt Ltd and Tepsol Saur Urja Pvt Ltd

Shri Sandeep Anand

Tepsol Sun Sparkle Pvt Ltd and Tepsol Saur Urja Pvt Ltd

Shri Bhagwat Singh

Malur Renewables Private Limited

Shri Mohit Jain

ReNew Dinkar Urja Private Limited

Shri Vishnu Khandelwal

Hindustan Zinc Limited

Shri Sumit Kaushik

Sunsure Energy Private Limited

Smt Poorva Pitke

Sprng Energy Private Limited

Shri Sourav Popli

Juniper Green Energy Pvt. Ltd.

Shri Keshav prasad

Bhadla Three SKP Green Ventures Private Limited

Shri Namit Jain

Abu Renewables India Pvt Ltd

Shri Yogesh Sanklecha

Acme Solar Holding Private Limited

Shri Ajay Pradhan

Acme Solar Holding Private Limited

Shri Ananth Raghavendra

Seven Renewable Power Pvt Ltd

Shri K A Vishwanath

XI Xergi Power Private Limited

Shri Kumar Vipul

Frugal Energy Private Limited

Shri Atul Bachikar

Luceo Solar Private Limited

Shri Santosh Narayan

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